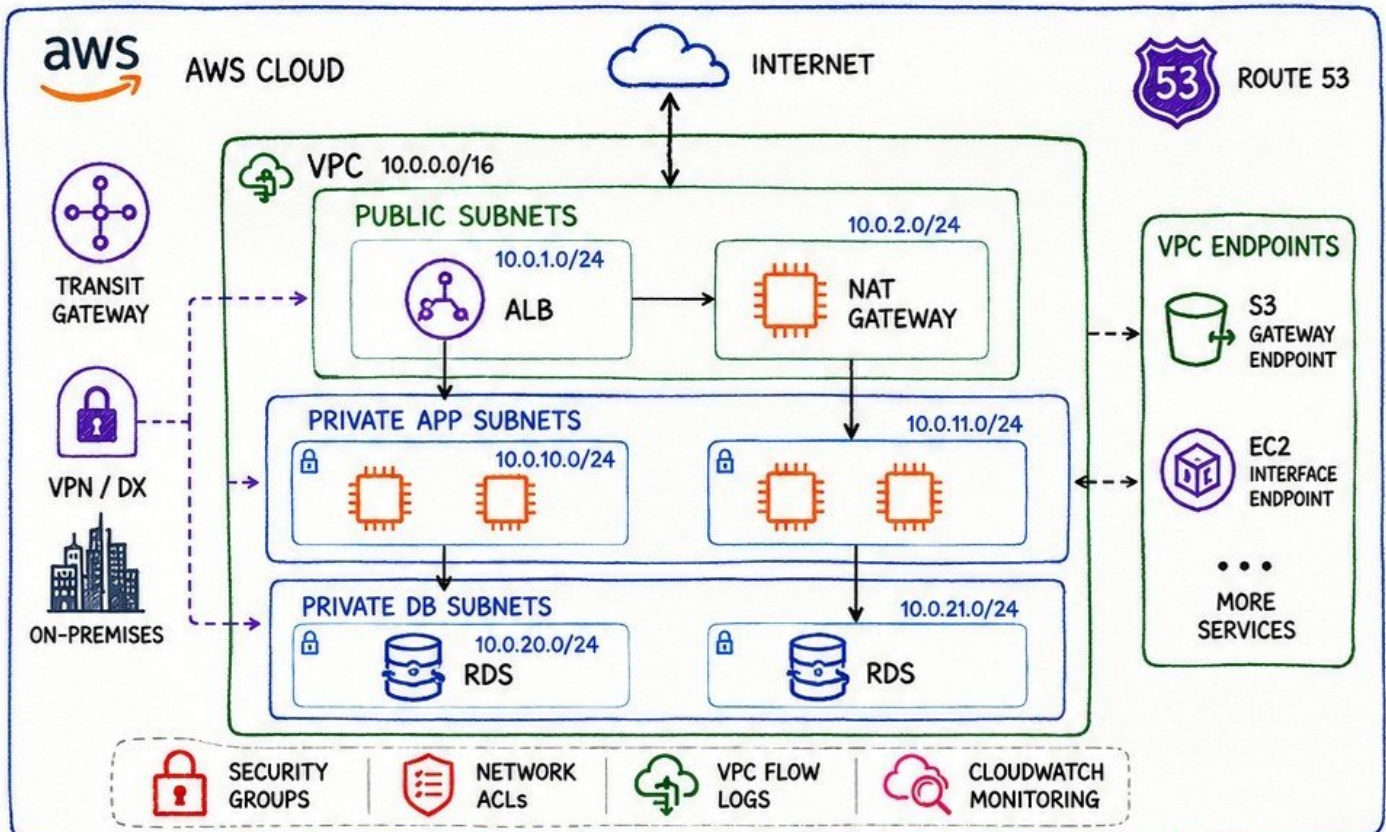


AWS NETWORKING HANDBOOK

20
PAGESFROM VPC FUNDAMENTALS TO
PRODUCTION NETWORK ARCHITECTUREVPC
FUNDAMENTALSSECURITY
BEST PRACTICESROUTING &
CONNECTIVITYHYBRID & PRIVATE
CONNECTIVITYMONITORING &
TROUBLESHOOTINGPRODUCTION
CHECKLISTS**DESIGN. SECURE. CONNECT.
MONITOR. SCALE.**Build resilient, secure, and
cost-effective AWS networks.

- ✓ Real-World Architectures
- ✓ Best Practices & Patterns
- ✓ Troubleshooting Guides
- ✓ Production Checklists

PERFECT FOR

- ✓ Cloud Engineers
- ✓ DevOps Engineers
- ✓ Solutions Architects
- ✓ SREs & Platform Engineers





1. AWS NETWORKING FOUNDATIONS

THE BASICS OF AWS NETWORKING YOU MUST MASTER

AWS GLOBAL NETWORKING MODEL



AWS operates on a **GLOBAL NETWORK** built for high availability, security, and scalability.



Every AWS Region is **PHYSICALLY ISOLATED** and connected to the **AWS GLOBAL BACKBONE**.



Traffic between Regions travels over the **AWS PRIVATE GLOBAL NETWORK**, not the public internet.



You design your **NETWORK INSIDE A REGION** using **AMAZON VPC**.

REGIONS AND AVAILABILITY ZONES

REGION



us-east-1
(N. Virginia)

AZ 1

us-east-1a



AZ 2

us-east-1b



AZ 3

us-east-1c



- A Region is a physical location in the world.
- Each Region has multiple isolated Availability Zones (AZs).
- AZs are connected with low-latency, high-bandwidth links.
- Design for high availability across AZs.

VPC FUNDAMENTALS



VPC

- Amazon VPC lets you launch AWS resources in a logically isolated network that you define.
- You choose your IP address range (CIDR block).
- You create subnets, route tables, gateways, and security controls.
- You have full control over your networking environment.
- Your VPC is isolated from other VPCs by default.

PUBLIC VS PRIVATE NETWORKING



PUBLIC NETWORKING

- ✓ Resources can be reached from the internet.
- ✓ Occurs in **PUBLIC SUBNETS**.
- ✓ Requires an **INTERNET GATEWAY (IGW)**.
- ✓ Uses **PUBLIC IP ADDRESSES** or **ELASTIC IPs**.
- ✓ Example: Web servers, ALBs, bastion hosts.

INTERNET



IGW



PUBLIC SUBNET



EC2
(Public IP)



PRIVATE NETWORKING

- ✓ Resources are not directly accessible from the internet.
- ✓ Occurs in **PRIVATE SUBNETS**.
- ✓ Uses **PRIVATE IP ADDRESSES**.
- ✓ Access to internet is through **NAT GATEWAY** (if needed).
- ✓ Example: App servers, databases, internal services.

PRIVATE SUBNET



EC2 / RDS
(Private IP)

NAT GATEWAY



INTERNET



HOW TRAFFIC MOVES THROUGH AWS

1

USER



Users send requests from their devices.

2

INTERNET



Traffic travels over the public internet.

3

ROUTE 53 / DNS



DNS resolves the domain name to an IP address.

4

EDGE LOCATION / CLOUDFRONT (OPTIONAL)



Traffic may pass through AWS edge locations.

5

ALB / IGW



Traffic reaches your ALB, IGW, or other entry point.

6

VPC



Traffic enters your VPC.

7

SUBNET & RESOURCES



Traffic is routed to subnets and reaches your resources.



- ✓ AWS provides a global network built for performance and security.
- ✓ Regions contain multiple AZs for fault tolerance.
- ✓ VPC is your isolated network in the cloud.
- ✓ Public networking is internet-facing; private networking is internal.
- ✓ Traffic follows a path: User → DNS → Edge → Entry → VPC → Resource.



IMPORTANT

- Always plan your network before launching resources.
- Use least privilege for both security and routing.
- Design for scalability, high availability, and security.





2. IPV4, IPV6, AND CIDR PLANNING

— Smart IP planning is the foundation of a secure, scalable, and production-ready AWS network. —

1. IPV4 ADDRESSING



- IPv4 uses 32-bit addresses.
- Format: 192.168.1.1
- Range: 0.0.0.0 – 255.255.255.255
- Total addresses: 4,294,967,296
- Not all addresses are usable in a subnet (network, broadcast).

2. PRIVATE IP RANGES



Use private IP ranges inside your VPC. They are not routable on the public internet.

PRIVATE IPV4 RANGES (RFC1918)

RANGE	CIDR
10.0.0.0 – 10.255.255.255	10.0.0.0/8
172.16.0.0 – 172.31.255.255	172.16.0.0/12
192.168.0.0 – 192.168.255.255	192.168.0.0/16

3. CIDR NOTATION



- CIDR (Classless Inter-Domain Routing) removes the old class system.
- Format: IP address / prefix length
- Example: 10.0.1.0/24
- /24 means the first 24 bits are the network portion.
- More bits in the prefix = smaller network.

4. SUBNET CALCULATIONS (IPV4)



EXAMPLE:

You have 10.0.0.0/16 and need subnets. How many subnets and hosts?

CIDR	SUBNET MASK	USABLE IPS	SUBNETS FROM /16
/16	255.255.0.0	65,534	1 (default VPC)
/20	255.255.240.0	4,094	16
/24	255.255.255.0	254	256
/26	255.255.255.192	62	1,024
/28	255.255.255.240	14	4,096

QUICK FORMULA

Usable Hosts = $2^{(32 - \text{prefix})} - 2$
(Subtract network address and broadcast address)

5. IPV6 BASICS



IPV6 ADDRESS FORMAT

2001:0db8:85a3:0000:0000:8a2e:0370:7334

- IPv6 uses 128-bit addresses.
- Virtually unlimited address space.
- No need for NAT.
- Built-in security (IPsec support).
- Example CIDR: 2001:db8::/32
- Use /64 for each subnet in AWS.

IPV6 SUBNET SIZE (BEST PRACTICE)

Use /64 for subnets.

Example:

- VPC CIDR: 2001:db8:1234::/56
- Subnet: 2001:db8:1234:1::/64

6. AVOIDING OVERLAPPING CIDRS

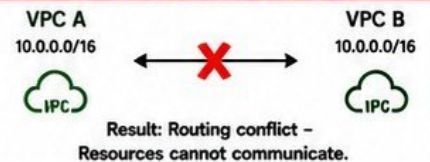


Overlapping CIDRs cause routing conflicts and connectivity failures.

AVOID OVERLAP BETWEEN:

- ✗ VPCs in the same account
- ✗ VPCs across peered connections
- ✗ On-premises networks
- ✗ Transit Gateway attachments
- ✗ VPN / Direct Connect networks

EXAMPLE OF BAD DESIGN



7. PRODUCTION IP PLANNING



- ✓ PLAN IP SPACE BEFORE YOU BUILD.
- ✓ USE PRIVATE IP RANGES ONLY.
- ✓ LEAVE ROOM TO GROW.
- ✓ SEGMENT BY PURPOSE, NOT BY ENVIRONMENT.
- ✓ ONE VPC CIDR SHOULD BE BIG ENOUGH TO LAST.
- ✓ DOCUMENT YOUR IP PLAN.
- ✓ REVIEW BEFORE PEERING OR CONNECTING.

EXAMPLE: PRODUCTION IP PLAN

COMPONENT	IPV4 CIDR	IPV6 CIDR	PURPOSE
VPC	10.0.0.0/16	2001:db8:1000::/56	Overall VPC range
Public Subnet 1 (AZ1)	10.0.1.0/24	2001:db8:1000:1::/64	ALB, NAT Gateway
Public Subnet 2 (AZ2)	10.0.2.0/24	2001:db8:1000:2::/64	ALB, NAT Gateway
Private App Subnet 1	10.0.10.0/24	2001:db8:1000:10::/64	Application Servers
Private App Subnet 2	10.0.11.0/24	2001:db8:1000:11::/64	Application Servers
Private DB Subnet 1	10.0.20.0/24	2001:db8:1000:20::/64	Database (AZ1)
Private DB Subnet 2	10.0.21.0/24	2001:db8:1000:21::/64	Database (AZ2)



KEY TAKEAWAY

Good IP planning prevents rework, reduces risk, and makes your network easy to scale.



REMEMBER

You cannot change the CIDR block of a VPC. Plan carefully from day one.



3. AMAZON VPC ARCHITECTURE

Build isolated, secure, and scalable networks in the AWS Cloud.

1. VPC COMPONENTS



VPC

Your virtual network in AWS.



SUBNETS

Divide your VPC into smaller IP ranges in each AZ.



ROUTE TABLES

Control traffic flow within your VPC.



INTERNET GATEWAY

Enables communication between your VPC and the internet.



NAT GATEWAY

Allows instances in private subnets to access the internet.



SECURITY GROUPS

Stateful firewall for instances.



NETWORK ACLS

Stateless firewall for subnets.



VPC ENDPOINTS

Private connectivity to AWS services without internet.

2. VPC CIDR BLOCKS

- Every VPC must have a CIDR block (IPv4).
- You cannot change the CIDR block after the VPC is created.
- Supported CIDR block range: /16 to /28 (for IPv4).

RECOMMENDED CIDR STRATEGY

VPC TYPE	CIDR BLOCK EXAMPLE	NOTES
Small	10.0.0.0/16	Up to 256 subnets
Medium	10.1.0.0/16	Good for most apps
Large	10.2.0.0/16	Many accounts and services
Enterprise	10.0.0.0/12	Multiple VPCs and growth

BEST PRACTICES

- ✓ Plan for growth.
- ✓ Use non-overlapping CIDRs.
- ✓ Leave space for expansion.
- ✓ Document your IP plan.

3. DEFAULT VS CUSTOM VPC



DEFAULT VPC

- Created automatically in every region.
- CIDR block: 172.31.0.0/16
- Has default subnets, route table, IGW, and SG.
- Good for quick testing only.



CUSTOM VPC

- You define your CIDR block.
- You create subnets, route tables, and gateways.
- Full control and better security.
- Required for production workloads.



ALWAYS USE CUSTOM VPC FOR PRODUCTION

Default VPC is shared and not isolated.

4. DNS SUPPORT



DNS

- DNS support enables the Amazon provided DNS server (Route 53 Resolver).
- It allows instances to resolve public DNS hostnames.
- Recommended: Enable DNS support for all VPCs.



ENABLE DNS SUPPORT

Allows Amazon DNS server at 169.254.169.253 to resolve names.

5. DNS HOSTNAMES



- DNS hostnames allow instances to get a public DNS name.

- Required for internet-facing resources (ALB, NAT Gateway, etc.).
- Without it, instances only have private IPs.



ENABLE DNS HOSTNAMES

Instances receive public DNS names that resolve to public IPs.

6. VPC DESIGN DECISIONS

- ✓ **IP ADDRESS RANGE**
How much IP space do you need now and in the future?
- ✓ **NUMBER OF AVAILABILITY ZONES**
How many AZs will you use for high availability?
- ✓ **SUBNET DESIGN**
How will you segment public, private app, and database subnets?
- ✓ **INTERNET ACCESS**
Which subnets need internet access? Use IGW or NAT Gateway?
- ✓ **SECURITY STRATEGY**
How will you use SGs, NACLs, and endpoints for security?
- ✓ **CONNECTIVITY REQUIREMENTS**
Do you need VPN, Direct Connect, Peering, or Transit Gateway?

TYPICAL VPC ARCHITECTURE OVERVIEW



KEY TAKEAWAY

A well-designed VPC is the foundation of a secure, scalable, and high-performing AWS network.



REMEMBER

Plan your IP space, design your subnets, enable DNS correctly, and choose custom VPCs for production.



4. PUBLIC AND PRIVATE SUBNETS

THE BUILDING BLOCKS OF A SECURE, SCALABLE, AND HIGHLY AVAILABLE VPC

1. SUBNET ARCHITECTURE



Subnets divide your VPC CIDR block into smaller IP ranges.



Each subnet exists in only ONE Availability Zone.



Each subnet is associated with a route table.



Use subnets to isolate tiers and control access.



Design subnets for security, performance, and scalability.

2. AVAILABILITY ZONE PLACEMENT

- An AWS Region has multiple AZs.
- Subnets must be created in an AZ.
- Spread subnets across multiple AZs for high availability.
- Never deploy all critical resources in a single AZ.

EXAMPLE: REGION WITH 3 AZs



3. PUBLIC SUBNETS



PUBLIC SUBNETS HAVE ROUTES TO THE INTERNET.

- ✓ Associated route table has a route to an Internet Gateway (0.0.0.0/0).
- ✓ Resources can have public IPs.
- ✓ Used for internet-facing resources (ALB, Bastion, NAT Gateway, etc.).
- ✓ Accessible directly from the internet.

EXAMPLE RESOURCES



ALB



BASTION
HOST



NAT
GATEWAY

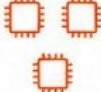
4. PRIVATE APPLICATION SUBNETS



PRIVATE SUBNETS WITHOUT DIRECT INTERNET ACCESS.

- ✓ No route to Internet Gateway.
- ✓ Instances have private IPs.
- ✓ Used for application servers, APIs, microservices, etc.
- ✓ Outbound internet access is through NAT Gateway.

EXAMPLE RESOURCES



EC2 INSTANCES
APPLICATION SERVERS
API SERVERS

5. PRIVATE DATABASE SUBNETS



MOST RESTRICTED SUBNETS.

- ✓ No direct internet access.
- ✓ No public IPs.
- ✓ Used for databases, caches, queues, etc.
- ✓ Only accessed by application tier.

EXAMPLE RESOURCES



RDS



ELASTICACHE
(REDIS / MEMCACHED)



AMAZON MQ

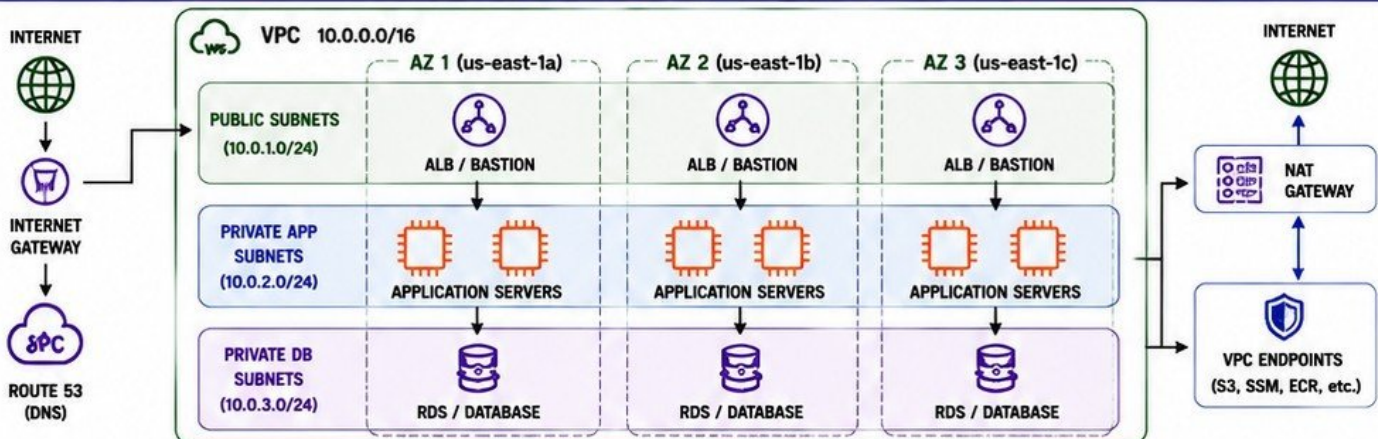
6. MULTI-AZ SUBNET DESIGN



DISTRIBUTE SUBNETS ACROSS MULTIPLE AZs FOR HIGH AVAILABILITY.

- ✓ Public, Private App, and Private DB subnets in each AZ.
- ✓ Allows your application to survive an AZ failure.
- ✓ Balance resources and traffic across AZs.
- ✓ Follow the principle of least privilege.

7. EXAMPLE: MULTI-AZ 3-TIER VPC ARCHITECTURE



KEY TAKEAWAYS



- ✓ Use subnets to segment tiers and control access.
- ✓ Place subnets across multiple AZs for resilience.
- ✓ Public subnets: internet-facing resources.
- ✓ Private App subnets: application workloads.
- ✓ Private DB subnets: data layer protection.
- ✓ Design for security, scalability, and availability.

BEST PRACTICES



- ✓ At least 2 AZs, 3 AZs recommended.
- ✓ One subnet per tier per AZ.
- ✓ No direct access from internet to private tiers.
- ✓ Use NAT Gateway for outbound internet from private subnets.
- ✓ Use VPC Endpoints to access AWS services privately.
- ✓ Plan IP ranges to avoid overlap and allow growth.





5. ROUTE TABLES AND TRAFFIC ROUTING

ROUTE TABLES CONTROL HOW TRAFFIC MOVES INSIDE AND OUTSIDE YOUR VPC

1. MAIN ROUTE TABLE



Every VPC has a **MAIN ROUTE TABLE** that is created automatically.

- ✓ You cannot delete the main route table.
- ✓ It is associated with all subnets by default.
- ✓ You can edit its routes.
- ✓ New subnets are associated with the main route table by default.

2. CUSTOM ROUTE TABLES



You can create **CUSTOM ROUTE TABLES** for more control.

- ✓ Associate specific subnets.
- ✓ Define custom routes.
- ✓ Isolate traffic between tiers.
- ✓ Support complex architectures and multiple exit points.

3. LOCAL ROUTES



Every route table has a **LOCAL ROUTE** automatically.

- ✓ Destination: Your VPC CIDR block (e.g., 10.0.0.0/16)
- ✓ Target: local
- ✓ This route cannot be deleted.
- ✓ It enables communication within your VPC.

4. ROUTE ASSOCIATION



A route table must be **ASSOCIATED** with subnets to be used.

- ✓ One subnet can have only one route table.
- ✓ A route table can be associated with multiple subnets.
- ✓ Use the main route table for simple setups.
- ✓ Use custom route tables for segmentation and control.

5. LONGEST PREFIX MATCHING

AWS uses **LONGEST PREFIX MATCHING** to determine the best route for a destination.



HOW IT WORKS

AWS looks for the most specific route (the longest matching CIDR prefix).

EXAMPLE ROUTES

DESTINATION (CIDR)	TARGET
10.0.0.0/16	local
10.0.1.0/24	eni-0abc123
0.0.0.0/0	igw-0def456

TRAFFIC TO 10.0.1.55

Matches 10.0.1.0/24 (more specific)
→ Goes to eni-0abc123

6. ROUTE PRIORITY

Route priority is determined by:

- LONGEST PREFIX MATCH**
More specific CIDR wins.
- STATIC BEFORE DYNAMIC**
Static routes > Propagated routes.
- BLACKHOLE ROUTES**
If a route target is unavailable, route becomes blackhole.

PRIORITY EXAMPLE

10.0.1.0/24
(Most Specific)
↓
10.0.0.0/16
↓
0.0.0.0/0
(Least Specific)

7. TROUBLESHOOTING ROUTING FAILURES

COMMON SYMPTOMS

- ✗ Instances cannot reach the internet.
- ✗ Cannot connect to services in other subnets.
- ✗ Time out or no response from destination.
- ✗ Intermittent connectivity issues.

TROUBLESHOOTING CHECKLIST

1

CHECK SUBNET ASSOCIATION

Ensure the subnet is associated with the correct route table.



2

CHECK ROUTE TABLE ENTRIES

Verify the route exists for the destination. Check CIDR and target.



3

VERIFY TARGET HEALTH

Ensure the target (IGW, NAT Gateway, TGW, VGW, etc.) is available.



4

CHECK SECURITY GROUPS & NACLs

Rules may be blocking the traffic.



5

TEST CONNECTIVITY

Use tools to test connectivity.

- ping
- tracer
- curl
- nc / telnet



COMMON ROUTING TARGETS



LOCAL

Used for routes inside the VPC.



IGW

Internet Gateway (Internet access).



NAT GATEWAY

Outbound internet for private subnets.



VGW

Virtual Private Gateway (VPN).



TGW

Transit Gateway (Hub connectivity).



VPC PEERING

Connects two VPCs privately.



ENDPOINT

Private access to AWS services.

KEY TAKEAWAYS



- ✓ Route tables control the flow of traffic in your VPC.
- ✓ Longest prefix matching decides the best route.
- ✓ Always associate the right route table with the right subnet.
- ✓ Validate routes and targets during design and troubleshooting.

REMEMBER



- ✓ There is always a local route for your VPC CIDR.
- ✓ 0.0.0.0/0 is used for internet or default routing.
- ✓ Use custom route tables for control and segmentation.
- ✓ Plan routes carefully to avoid blackholes and downtime.



6. INTERNET GATEWAY AND PUBLIC CONNECTIVITY

MAKE YOUR VPC REACHABLE FROM THE INTERNET THE RIGHT WAY

1. INTERNET GATEWAY



An Internet Gateway (IGW) enables communication between your VPC and the internet.

- ✓ IGW is a horizontal, redundant, and highly available AWS service.
- ✓ It is attached to your VPC.
- ✓ It supports IPv4 traffic only.
- ✓ One VPC can have one IGW attached at a time.

2. PUBLIC IPv4 ADDRESSES



54.239.10.25

Public IPv4 addresses are used to communicate directly over the internet.

- ✓ Required for internet-facing resources.
- ✓ Automatically assigned to EC2 instances in a public subnet (if enabled).
- ✓ Released when the instance is stopped or terminated (ephemeral).
- ✓ You pay for public IPv4 addresses that are not in use.

3. ELASTIC IP ADDRESSES

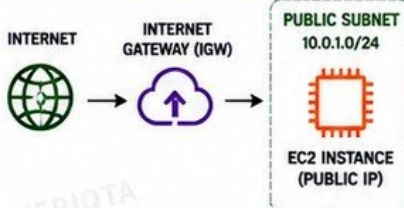


18.207.45.10

Elastic IPs (EIPs) are static, public IPv4 addresses that you allocate and control.

- ✓ Static and persistent until you release them.
- ✓ Can be re-associated with another instance.
- ✓ Useful for production workloads.
- ✓ Charged when allocated, whether in use or not.

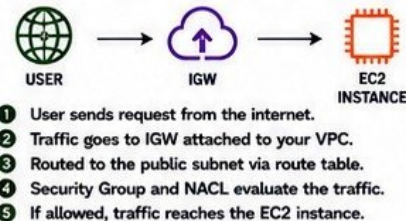
4. PUBLIC EC2 CONNECTIVITY



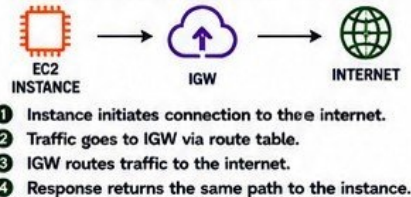
- ✓ EC2 instance must be in a PUBLIC SUBNET.
- ✓ Subnet route table must have a route to IGW (0.0.0.0/0 → IGW).
- ✓ Instance must have a PUBLIC IPv4 or EIP.
- ✓ Security Group must allow inbound traffic.
- ✓ Network ACL must allow the traffic.

5. INBOUND AND OUTBOUND TRAFFIC

INBOUND TRAFFIC (INTERNET → INSTANCE)



OUTBOUND TRAFFIC (INSTANCE → INTERNET)



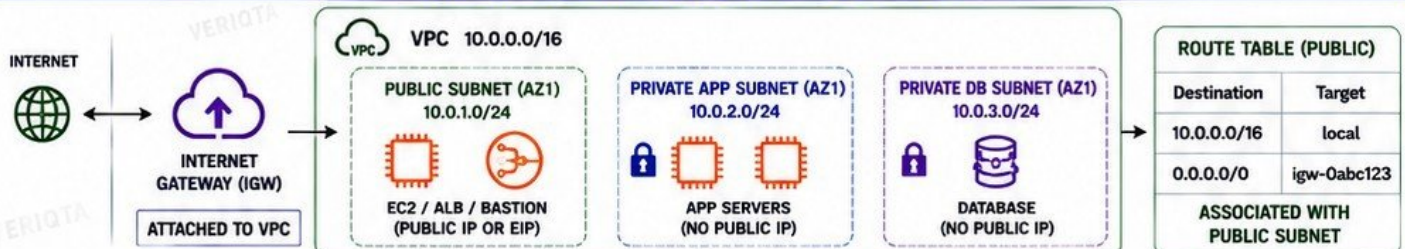
6. BUILDING A TRUE PUBLIC SUBNET

- ✓ Subnet has a route to an Internet Gateway.
- ✓ Resources in the subnet can obtain public IPs.
- ✓ Instances can communicate with the internet.
- ✓ Used for internet-facing resources like ALB, Bastion, NAT Gateway, Web Servers.

EXAMPLE: PUBLIC SUBNET SETUP

1	VPC CIDR BLOCK	10.0.0.0/16
2	PUBLIC SUBNET (AZ1)	10.0.1.0/24
3	ROUTE TABLE	
	Destination:	0.0.0.0/0
	Target:	igw-0abc123def4567890
4	ASSOCIATION	Public Subnet → Public Route Table
5	IGW ATTACHED TO VPC	Yes
6	AUTO-ASSIGN PUBLIC IP	Enabled

TRUE PUBLIC SUBNET ARCHITECTURE EXAMPLE



KEY TAKEAWAYS

- ✓ Attach an Internet Gateway to your VPC for internet access.
- ✓ Use public subnets with a route to IGW for internet-facing resources.
- ✓ Assign public IPs or use Elastic IPs for stable connectivity.
- ✓ Security Groups and NACLs control the traffic flow.
- ✓ Design your public subnet correctly for a secure and reliable setup.



REMEMBER

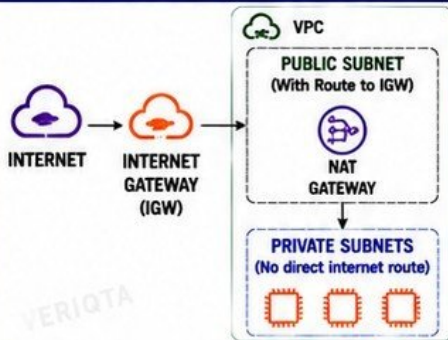
- ✓ IGW enables internet connectivity for your VPC.
- ✓ Public subnets + route to IGW = internet access.
- ✓ Private subnets do NOT have direct internet access.
- ✓ Use EIP for production systems that require stable public IPs.
- ✓ Always follow least privilege in Security Groups and NACLs.



7. NAT GATEWAY AND PRIVATE INTERNET ACCESS

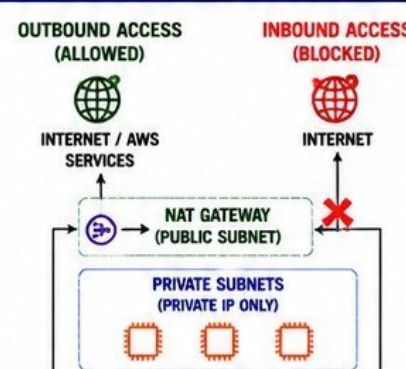
ENABLE SECURE OUTBOUND INTERNET ACCESS FOR PRIVATE RESOURCES

1. NAT GATEWAY ARCHITECTURE



- ✓ Private instances do not have public IPs.
- ✓ They send outbound traffic to the NAT Gateway.
- ✓ NAT Gateway has a route to the Internet Gateway.
- ✓ Response traffic returns through the NAT Gateway.
- ✓ Inbound traffic from the internet is NOT allowed.

2. PRIVATE SUBNET OUTBOUND ACCESS



- 1 Instances initiate connection to the internet.
- 2 Traffic goes to NAT Gateway.
- 3 NAT Gateway routes traffic to the internet.
- 4 Response comes back to NAT Gateway.
- 5 NAT Gateway forwards response to instance.

3. ROUTE CONFIGURATION

PUBLIC SUBNET ROUTE TABLE

DESTINATION	TARGET
10.0.0.0/16	local
0.0.0.0/0	igw-0abc1234567890

PRIVATE SUBNET ROUTE TABLE

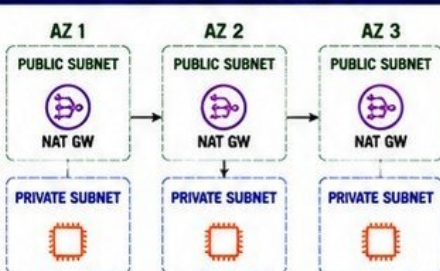
DESTINATION	TARGET
10.0.0.0/16	local
0.0.0.0/0	nat-0abc1234567890



KEY POINT

Private subnets must have 0.0.0.0/0 route pointing to NAT Gateway for internet access.

4. MULTI-AZ NAT DESIGN



- ✓ Deploy one NAT Gateway in each AZ.
- ✓ Private subnets in each AZ use the NAT Gateway in the same AZ.
- ✓ Ensures high availability and removes single points of failure.
- ✓ Cross-AZ traffic is avoided (better performance and lower cost).

5. NAT GATEWAY FAILURE SCENARIOS

- ✗ **NAT GATEWAY FAILURE**
Instances in that AZ lose internet access.
- ✗ **IGW DETACHMENT**
No internet connectivity for NAT Gateway.
- ✗ **ROUTE TABLE MISCONFIGURATION**
Traffic goes to blackhole or wrong target.
- ✗ **EIP RELEASED (IF ASSOCIATED)**
NAT Gateway gets a new IP (if EIP not used).
- ✗ **AZ OUTAGE**
If only one NAT Gateway exists, that AZ is effectively isolated from the internet.



BEST PRACTICE

Always deploy NAT Gateway in each AZ and monitor its health.

6. COST CONSIDERATIONS



NAT GATEWAY COSTS

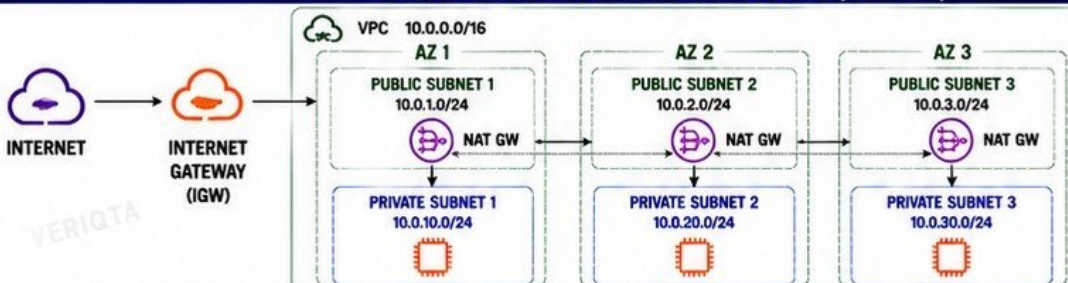
- ✓ Hourly charge per NAT Gateway.
- ✓ Data processing charge per GB.
- ✓ Cross-AZ data transfer charges apply if traffic goes to NAT in another AZ.
- ✓ Each AZ requires its own NAT Gateway for high availability (increases cost).



COST OPTIMIZATION TIPS

- ✓ Use one NAT Gateway per AZ, not a single NAT for all AZs.
- ✓ Avoid cross-AZ traffic.
- ✓ Release unused NAT Gateways.
- ✓ Use VPC Endpoints for AWS services to avoid NAT charges.

7. PRIVATE INTERNET ACCESS ARCHITECTURE (MULTI-AZ)



PRIVATE INSTANCES CAN:

- ✓ Access the internet.
- ✓ Download updates.
- ✓ Pull container images.
- ✓ Access external APIs.
- ✓ Reach AWS services via NAT Gateway.



KEY TAKEAWAYS

- ✓ NAT Gateway provides outbound-only internet access for private subnets.
- ✓ Place NAT Gateway in a public subnet with a route to IGW.
- ✓ Configure private subnet route table to use NAT Gateway.
- ✓ Deploy one NAT Gateway per AZ for high availability.
- ✓ Monitor and plan for cost, failures, and best practices.



REMEMBER

- ✓ NAT Gateway does NOT allow inbound connections.
- ✓ Instances in private subnets remain private.
- ✓ Always check route tables and NACLs when troubleshooting.
- ✓ Use CloudWatch for NAT Gateway monitoring.
- ✓ Design for resilience, performance, and cost efficiency.



8. SECURITY GROUPS

STATEFUL FIREWALLS THAT CONTROL TRAFFIC TO YOUR RESOURCES

1. STATEFUL FILTERING



Security Groups are stateful.

- ✓ If you allow inbound traffic, the response is allowed automatically.
- ✓ You only need to define rules for the inbound direction.
- ✓ Return traffic for outbound requests is automatically allowed.
- ✓ No explicit rules are needed for response traffic.

2. INBOUND RULES



Control traffic coming into your resources.

- ✓ Specify: Type, Protocol, Port Range, Source
- ✓ Source can be: IP, CIDR block, or another SG
- ✓ Use the principle of least privilege
- ✓ Only allow what is required

EXAMPLE INBOUND RULES

Type	Protocol	Port	Source
HTTPS	TCP	443	0.0.0.0/0
SSH	TCP	22	10.0.1.0/24
HTTP	TCP	80	203.0.113.0/24

3. OUTBOUND RULES



Control traffic leaving your resources.

- ✓ Default outbound rule allows all traffic.
- ✓ Restrict outbound to only what is necessary.
- ✓ Use specific destinations, ports, and protocols.
- ✓ Improves security and reduces risk.

EXAMPLE OUTBOUND RULES

Type	Protocol	Port	Destination
HTTPS	TCP	443	0.0.0.0/0
DNS	UDP	53	0.0.0.0/0
Custom	TCP	3306	10.0.2.0/24

4. SECURITY GROUP REFERENCES



Use Security Groups as sources/destinations.

- ✓ Allows you to reference one SG from another.
- ✓ Automatically adapts to IP changes.
- ✓ Simplifies rules and improves security.
- ✓ Recommended for communication between AWS resources.

EXAMPLE

Source: sg-app-tier
(Allows traffic only from instances in sg-app-tier)

5. APPLICATION-TIER ISOLATION (EXAMPLE ARCHITECTURE)

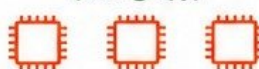
INTERNET



LOAD BALANCER
(SG: sg-alb)



APPLICATION TIER
(SG: sg-app)



DATABASE TIER
(SG: sg-db)



sg-alb (PUBLIC)

INBOUND RULES

Type	Protocol	Port	Source
HTTP	TCP	80	0.0.0.0/0
HTTPS	TCP	443	0.0.0.0/0

OUTBOUND RULES

Type	Protocol	Port	Destination
HTTP	TCP	80	sg-app
HTTPS	TCP	443	sg-app

sg-app (PRIVATE)

INBOUND RULES

Type	Protocol	Port	Source
HTTP	TCP	80	sg-alb
HTTPS	TCP	443	sg-alb

OUTBOUND RULES

Type	Protocol	Port	Destination
HTTP	TCP	80	0.0.0.0/0
HTTPS	TCP	443	0.0.0.0/0
Custom	TCP	3306	sg-db

sg-db (PRIVATE)

INBOUND RULES

Type	Protocol	Port	Source
MYSQL	TCP	3306	sg-app

OUTBOUND RULES

Type	Protocol	Port	Destination
All Traffic	All	All	(Response) Automated



TRAFFIC FLOW: INTERNET → ALB → APP TIER → DATABASE TIER



DATABASE NOT ACCESSIBLE FROM THE INTERNET

6. LEAST-PRIVILEGE DESIGN



- ✓ Allow only the traffic that is required.
- ✓ Limit access by IP, CIDR, or Security Group.
- ✓ Avoid wide-open rules (! 0.0.0.0/0) when possible.
- ✓ Review and audit rules regularly.
- ✓ Combine with NACLs for defense in depth.

7. COMMON SECURITY GROUP MISTAKES



- ✗ Allowing all inbound traffic (0.0.0.0/0) on all ports.
- ✗ Overly permissive outbound rules.
- ✗ Not using SG references between tiers.
- ✗ Placing application and database in the same SG.
- ✗ Not reviewing rules after deployments.
- ✗ Ignoring unused rules.
- ✗ Relying only on Security Groups (no NACLs).

SECURITY GROUP BEST PRACTICES



USE SG REFERENCES
Prefer SG-to-SG rules over IP addresses.



KEEP IT SIMPLE
Fewer rules are easier to manage and audit.



REVIEW REGULARLY
Remove unused rules and old temporary access.



DEFAULT DENY
Start with no access. Add only what is required.



DOCUMENT RULES
Document the purpose of each rule and its owner.



TEST CHANGES
Test thoroughly before applying to production.



REMEMBER: SECURITY GROUPS ARE STATEFUL, BUT YOU ARE RESPONSIBLE FOR DEFINING THE RIGHT RULES.
LEAST PRIVILEGE + REGULAR REVIEW = STRONG SECURITY.



9. NETWORK ACLs (NACLs)

SUBNET FIREWALLS THAT CONTROL TRAFFIC AT THE BOUNDARY

1. STATELESS FILTERING



Network ACLs are stateless.

- ✓ Each rule is evaluated independently.
- ✓ A return response must be explicitly allowed.
- ✓ NACLs evaluate traffic before it reaches Security Groups.

2. INBOUND AND OUTBOUND RULES

INBOUND

(From outside the subnet to resources inside)



- ✓ Evaluated first
- ✓ Checked in rule number order
- ✓ Must explicitly allow return traffic

OUTBOUND

(From resources inside the subnet to outside)



- ✓ Evaluated after inbound
- ✓ Checked in rule number order
- ✓ Must explicitly allow outbound traffic

3. RULE NUMBERING

- ✓ Rules are evaluated in ascending order (lowest to highest).
- ✓ The first rule that matches a request is applied.
- ✓ Use rule numbers with gaps (e.g., 100, 110, 120) for future changes.

EVALUATION ORDER

100

↓

110

↓

120

↓

...

↓

* (DENY ALL)

4. ALLOW AND DENY RULES

ALLOW RULE

- ✓ Allows matching traffic.

If no rule allows it, the traffic is denied.

DENY RULE

- ✗ Explicitly blocks matching traffic.

Use deny rules to block unwanted traffic early.

5. EPHEMERAL PORTS

Return traffic uses ephemeral (dynamic) ports.

COMMON EPHEMERAL PORT RANGES

Linux (Default)	32768 – 60999
Windows (Older)	1024 – 5000
Windows (Newer)	49152 – 65535

- ✓ Always allow ephemeral ports for return traffic in NACL outbound rules.

6. SECURITY GROUPS vs NACLs

FEATURE	SECURITY GROUPS	NACLs
Type	Stateful	Stateless
Applies To	ENI (Instance level)	Subnet level
Rule Direction	Implicit (Return allowed)	Explicit (Both directions)
Default Behavior	Allow all outbound	Deny all
Rule Evaluation	All rules evaluated	First match wins
Use Case	Workload-level security	Subnet-level firewall
Granularity	Higher	Lower

7. SAMPLE NACL CONFIGURATION (PUBLIC SUBNET)

INBOUND RULES

RULE #	TYPE	PROTOCOL	PORT RANGE	SOURCE	ACTION
100	ALLOW	TCP	22	0.0.0.0/0 (SSH)	ALLOW
110	ALLOW	TCP	80	0.0.0.0/0 (HTTP)	ALLOW
120	ALLOW	TCP	443	0.0.0.0/0 (HTTPS)	ALLOW
130	ALLOW	ALL	ALL	10.0.0.0/16 (VPC CIDR)	ALLOW
*	DENY	ALL	ALL	0.0.0.0/0	DENY

OUTBOUND RULES

RULE #	TYPE	PROTOCOL	PORT RANGE	SOURCE	ACTION
100	ALLOW	ALL	ALL	0.0.0.0/0	ALLOW
110	ALLOW	ALL	ALL	10.0.0.0/16	ALLOW
120	DENY	ALL	ALL	0.0.0.0/0	DENY
*	DENY	ALL	ALL	0.0.0.0/0	DENY

WHERE NACLs APPLY



VPC
10.0.0.0/16



PUBLIC SUBNET
10.0.1.0/24

EC2 INSTANCES

8. TROUBLESHOOTING NACL FAILURES

1. VERIFY SUBNET NACL ASSOCIATION



Ensure the subnet is associated with the correct NACL.

2. CHECK RULE LOGIC



Ensure rules are in the correct order and the first match allows the traffic.

3. VERIFY BOTH DIRECTIONS



Remember NACLs are stateless. Allow both inbound and outbound traffic.

4. CHECK RETURN TRAFFIC



Allow ephemeral ports (1024–65535) in outbound rules.

5. CHECK FOR DENY RULES



A deny rule placed above an allow rule will block the traffic.

6. USE VPC FLOW LOGS FOR VISIBILITY



Enable Flow Logs and look for REJECT entries in the logs.

KEY TAKEAWAYS

- ✓ NACLs are stateless and subnet-level firewalls.
- ✓ Rules are processed in order; first match wins.
- ✓ Allow both inbound and outbound for return traffic.
- ✓ Use ephemeral port ranges for outbound rules.
- ✓ Combine NACLs and Security Groups for defense in depth.

BEST PRACTICES

- ✓ Keep NACLs simple and focused on broad filtering.
- ✓ Allow only required CIDRs and ports.
- ✓ Use rule number gaps for future updates.
- ✓ Test changes in a non-production environment.
- ✓ Monitor with VPC Flow Logs and CloudWatch.



10. DNS AND AMAZON ROUTE 53

SCALABLE, HIGHLY AVAILABLE, AND RELIABLE DNS FOR YOUR APPLICATIONS

1. DNS FUNDAMENTALS



- ✓ DNS (Domain Name System) translates human-readable names to IP addresses.
- ✓ It is the phonebook of the internet.
- ✓ Route 53 is AWS' scalable and highly available DNS service.
- ✓ It answers queries using authoritative nameservers.

2. ROUTE 53 HOSTED ZONES



- ✓ A hosted zone is a container for records for a domain.
- ✓ Two types of hosted zones:
 - Public Hosted Zone (Internet-facing)
 - Private Hosted Zone (within VPC only)
- ✓ Each domain must have exactly one public hosted zone per AWS account.

EXAMPLE

Domain: example.com
Hosted Zone ID: Z2ABCEDEF123456
Name Servers:
ns-123.awsdns-45.com
ns-678.awsdns-90.net
ns-345.awsdns-67.org
ns-901.awsdns-23.co.uk

3. DNS RECORD TYPES (COMMON)

TYPE	DESCRIPTION	EXAMPLE
A	Maps domain to IPv4 address	example.com → 192.0.2.10
AAAA	Maps domain to IPv6 address	example.com → 2001:db8::10
CNAME	Alias one domain to another	www → example.com
MX	Mail exchange server	example.com → mail.example.com
NS	Name server record for delegated zone	example.com → ns-123.awsdns.com
TXT	Text records for verification (SPF, DKIM)	v=spf1 include:_spf...
SRV	Service location (priority, weight, port)	_sip._tcp.example.com
PTR	Reverse DNS (IP to domain)	10.2.0.5 → ip-10-2-0-5.ec2

4. ALIAS RECORDS



- ✓ Alias records let you point a domain to AWS resources.
- ✓ Works for: ELB, CloudFront, S3 website, API Gateway, VPC endpoints, etc.
- ✓ Alias records are zone apex (root domain) friendly.
- ✓ No additional cost for Route 53 queries.

EXAMPLE

Type: A - Alias
Name: example.com
Alias Target: dualstack.my-alb-123456.us-east-1.elb.amazonaws.com
Alias Hosted Zone ID: Z3SSXD0TRQ7X7K

5. TTL (TIME TO LIVE)



- ✓ TTL determines how long DNS responses are cached.
- ✓ Lower TTL = faster changes, higher DNS query volume.
- ✓ Higher TTL = less traffic, slower propagation.

COMMON TTL GUIDELINES

60 seconds	Rapidly changing records
300 seconds (5 min)	Most applications
3600 seconds (1 hour)	Stable records
86400 seconds (1 day)	Very stable records

6. ROUTING POLICIES

	SIMPLE	Default policy. Use for most domains.
	WEIGHTED	Split traffic based on weights (e.g., 80% / 20%).
	LATENCY	Route users to region with the lowest latency.
	FAILOVER	Active-passive setup for high availability.
	GEOLOCATION	Route based on user's geographic location.
	GEOPROXIMITY	Route based on AWS region proximity to users.
	MULTIVALUE ANSWER	Return multiple healthy records.

7. HEALTH CHECKS



- ✓ Monitor the health of your resources from multiple AWS regions.
- ✓ If a resource fails the check, Route 53 can route traffic away (with failover or multi-value policies).
- ✓ Supports HTTP, HTTPS, TCP, and calculated health checks.

EXAMPLE

Check Type: HTTPS
Endpoint: https://app.example.com/health
Frequency: 30 seconds
Failure Threshold: 3
Regions: us-east-1, eu-west-1, ap-south-1

8. PRIVATE HOSTED ZONES



- ✓ Private hosted zones are accessible only within your VPC.
- ✓ Used for internal domains like internal.example.local.
- ✓ Records are not resolvable from the internet.
- ✓ Require VPC association.



9. ROUTE 53 BEST PRACTICES



- ✓ Use meaningful, consistent domain names.
- ✓ Choose appropriate TTL values.
- ✓ Use Alias records for AWS resources.
- ✓ Enable DNSSEC for domain security.
- ✓ Use health checks with failover or multi-value answers.
- ✓ Regularly review records and remove unused entries.



KEY TAKEAWAY: ROUTE 53 IS MORE THAN DNS → IT IS A HIGHLY AVAILABLE, SCALABLE, AND COST-EFFECTIVE WAY TO ROUTE INTERNET TRAFFIC TO YOUR APPLICATIONS.





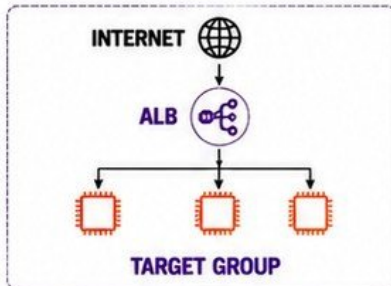
11. ELASTIC LOAD BALANCING

DISTRIBUTE TRAFFIC ACROSS MULTIPLE RESOURCES FOR HIGH AVAILABILITY AND SCALABILITY

1. APPLICATION LOAD BALANCER (ALB)



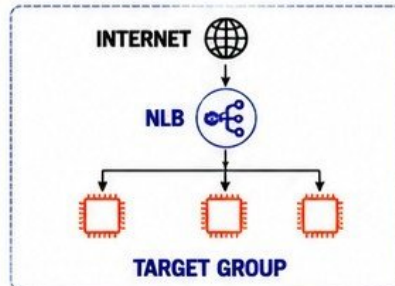
- ✓ Operates at Layer 7 (HTTP/HTTPS).
- ✓ Content-based routing (host, path, headers).
- ✓ Ideal for web applications and microservices.
- ✓ Supports HTTP/2, WebSockets, and gRPC.
- ✓ Provides advanced routing and observability.



2. NETWORK LOAD BALANCER (NLB)



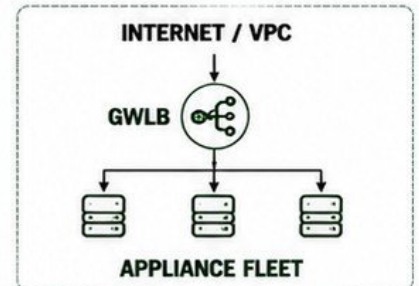
- ✓ Operates at Layer 4 (TCP/UDP/TLS).
- ✓ Ultra-high performance and low latency.
- ✓ Handles millions of requests per second.
- ✓ Preserves client IP address.
- ✓ Ideal for TCP-based applications.



3. GATEWAY LOAD BALANCER (GWLB)



- ✓ Operates at Layer 3 (IP packets).
- ✓ Used for third-party virtual appliances.
- ✓ Transparent to applications.
- ✓ Use cases: firewalls, IDS/IPS, DPI, etc.
- ✓ Scales and manages appliance fleets.



4. LISTENERS



- ✓ A listener checks for connection requests using the protocol and port you configure.
- ✓ Listeners forward requests to one or more target groups using rules.
- ✓ Default action or rule-based routing.

COMMON LISTENER CONFIGURATIONS

PROTOCOL	PORT
HTTP	80
HTTPS	443
HTTPS (gRPC)	443
TCP	Any (NLB)
TLS	Any (NLB)

5. TARGET GROUPS



- ✓ A target group is a collection of targets (instances, IPs, or Lambda).
- ✓ Requests from the load balancer are routed to the healthy targets in the target group.
- ✓ Supports multiple target types.

TARGET TYPES

TYPE	EC2 Instances
Instance	IP addresses from any VPC
IP	AWS Lambda functions
ALB	Another Application Load Balancer

6. HEALTH CHECKS



- ✓ Health checks determine the health of targets.
- ✓ Only healthy targets receive traffic.
- ✓ You can configure protocol, path, port, and thresholds.

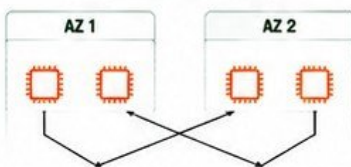
HEALTH CHECK SETTINGS

SETTING	DESCRIPTION
Protocol	HTTP, HTTPS, TCP
Port	Traffic port or override port
Path (HTTP/HTTPS)	e.g., /health, /status
Healthy Threshold	Consecutive successes
Unhealthy Threshold	Consecutive failures
Interval	Time between checks
Timeout	Time to wait for response

7. CROSS-ZONE TRAFFIC



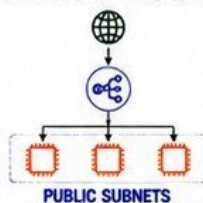
- ✓ When enabled, the load balancer routes traffic to healthy targets in all AZs.
- ✓ Improves availability and fault tolerance.
- ✓ Recommended: Keep it ENABLED (default for ALB and NLB).



8. INTERNAL vs INTERNET-FACING

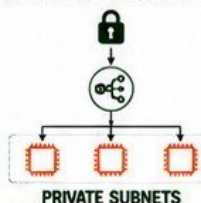
INTERNET-FACING

- ✓ Has a public DNS name.
- ✓ Accessible from the internet.
- ✓ Requires public subnets.
- ✓ Use for public-facing apps.



INTERNAL

- ✓ Has a private DNS name.
- ✓ Accessible only within the VPC.
- ✓ Requires private subnets.
- ✓ Use for internal applications.



9. LOAD BALANCER COMPARISON

FEATURE	ALB	NLB	GWLB
OSI LAYER	7	4	3
PROTOCOLS	HTTP, HTTPS, gRPC, WS	TCP, UDP, TLS	IP Packets
USE CASE	Web apps, Microservices	High performance	Network appliances
TARGET TYPES	Instance, IP, Lambda	Instance, IP	Appliances
CLIENT IP	Preserved	Preserved	Not preserved
STATIC IP	No	Yes	No
CONTENT-BASED ROUTING	Yes	No	No

10. ELASTIC LOAD BALANCING BEST PRACTICES



USE ALB FOR WEB APPLICATIONS.



USE NLB FOR HIGH PERFORMANCE AND LOW LATENCY.



ENABLE CROSS-ZONE TRAFFIC.



CONFIGURE HEALTH CHECKS PROPERLY.



USE INTERNAL LBs FOR PRIVATE APPS.



ORGANIZE TARGETS INTO LOGICAL TARGET GROUPS.



MONITOR WITH CLOUDWATCH METRICS.



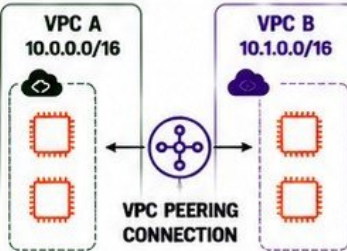
REMEMBER: A WELL-DESIGNED LOAD BALANCER IS KEY TO HIGH AVAILABILITY, SCALABILITY, AND PERFORMANCE.



12. VPC PEERING

PRIVATE NETWORK CONNECTIVITY BETWEEN VPCS

1. PEERING ARCHITECTURE



- Creates a network connection between two VPCs.
- Instances behave as if they are on the same network.
- Uses private IPv4 addresses.
- No gateway, no encryption, no bandwidth charge.

2. ROUTE CONFIGURATION

VPC A ROUTE TABLE

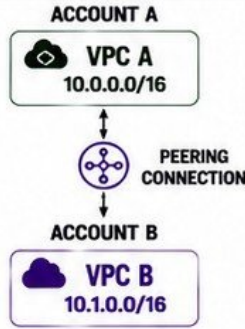
DESTINATION	TARGET
10.0.0.0/16	local
10.1.0.0/16	pcx-123abc
0.0.0.0/0	igw-aaaaaaa

VPC B ROUTE TABLE

DESTINATION	TARGET
10.1.0.0/16	local
10.0.0.0/16	pcx-123abc
0.0.0.0/0	igw-bbbbbb

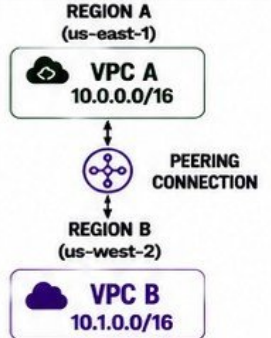
- Add a route to the peer CIDR.
- Target must be the Peering Connection ID (pcx-xxxx).
- Routes must be added on both sides (not automatic).

3. CROSS-ACCOUNT PEERING



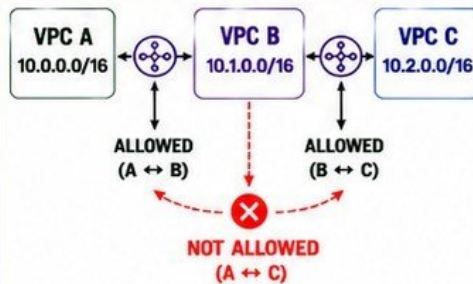
- Supported between VPCs in different AWS accounts.
- Peering connection must be accepted by the requester.
- Use Resource Access Manager (RAM) for management.

4. CROSS-REGION PEERING



- Supported between VPCs in different AWS Regions.
- Traffic stays on AWS backbone.
- Use correct Region for each peering connection.

5. NON-TRANSITIVE ROUTING



- Peering is NOT transitive.
- VPC A cannot reach VPC C through VPC B.
- Every VPC must have a direct peering connection.

6. CIDR OVERLAP LIMITATIONS



EXAMPLES

✓	10.0.0.0/16 ↔ 10.1.0.0/16	CAN PEER
✗	10.0.0.0/16 ↔ 10.0.0.0/16	CANNOT PEER
✗	10.0.0.0/16 ↔ 10.0.1.0/24	CANNOT PEER

Use non-overlapping CIDR ranges.

7. WHEN PEERING STOPS SCALING

- Too many VPCs (Full Mesh Peering) causes exponential connections.
- Difficult to manage routes and peering connections.
- Propagation delays and inconsistent routing.
- No centralized route control.
- Security and compliance become harder to enforce.

USE TRANSIT GATEWAY WHEN YOU HAVE:

- More than 3-5 VPCs
- Multiple accounts
- Multiple Regions
- Complex routing requirements



8. VPC PEERING BEST PRACTICES



PLAN CIDR RANGES EARLY

Avoid overlaps and redesign later.



UPDATE ROUTES ON BOTH SIDES

Traffic will not work until routes exist on both VPCs.



USE SECURITY GROUPS & NACLs

Peering does not add security. You must control access.



MONITOR TRAFFIC AND CONNECTIVITY

Use VPC Flow Logs, CloudWatch, and Reachability Analyzer.



KEEP PEERING SIMPLE

Limit connections and document all routes.



USE TRANSIT GATEWAY FOR SCALE

Centralized routing, more control, better scalability.



TAG AND DOCUMENT EVERY PEERING CONNECTION

Helps with visibility, billing, and audit.

VPC PEERING SUMMARY

FEATURE	SUPPORTED?	NOTES
Same Region Peering	✓ YES	Fully supported.
Cross-Region Peering	✓ YES	Supported across AWS Regions.
Cross-Account Peering	✓ YES	Supported. Must accept the request.
Overlapping CIDRs	✗ NO	Not allowed.
Transitive Routing	✗ NO	Peering is NOT transitive.
IPv6 Peering	✓ YES (With IPv6 CIDR)	Must enable IPv6 and add routes.
Encryption	✗ NO	Traffic is not encrypted (use VPN/TGW for encryption).
Bandwidth Charges	✓ NO	No charge for data transfer between peered VPCs.

QUICK CHECKLIST

- ✓ CIDRs are non-overlapping
- ✓ Peering connection created and accepted
- ✓ Routes added on both sides
- ✓ Security groups and NACLs configured
- ✓ Connectivity tested
- ✓ Monitoring and Flow Logs enabled

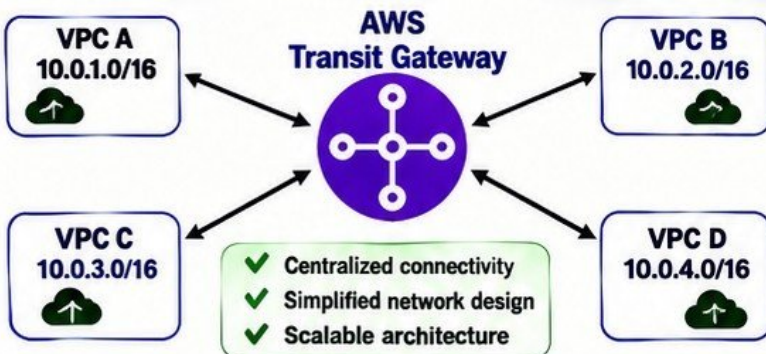




13. AWS TRANSIT GATEWAY

HUB-AND-SPOKE NETWORKING FOR MULTI-VPC AND MULTI-ACCOUNT

1. TRANSIT GATEWAY ARCHITECTURE (HUB-AND-SPOKE)



2. VPC ATTACHMENTS

ATTACHMENT TYPE	USE CASE	SUPPORT
VPC Attachment	Connect VPCs	Yes
VPN Attachment	Connect on-prem	Yes
Direct Connect	Private AWS link	Yes
Transit VPC	Network hub	Yes

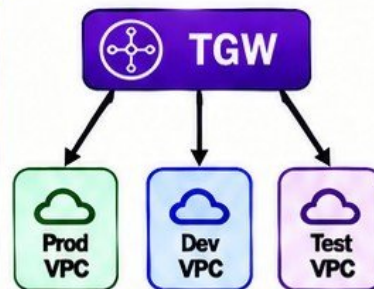
- ✓ Attach multiple VPCs
- ✓ Support multiple account networking
- ✓ Native AWS integration

3. TRANSIT GATEWAY ROUTE TABLES

ROUTE TABLE	PURPOSE
Default	Share routes with all attachments
Custom 1	Isolate production
Custom 2	Isolate dev/test
Custom 3	Business unit segmentation

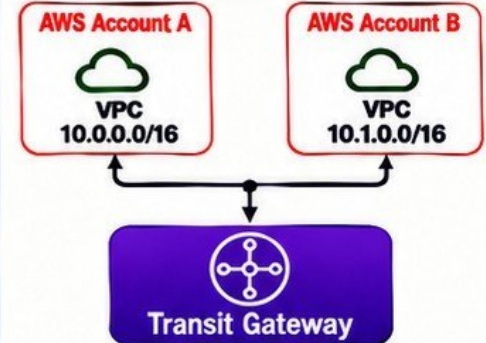
- ✓ Control traffic flow
- ✓ Segment environments
- ✓ Apply least-privilege routing

4. SEGMENTATION



- ✓ Isolate traffic by environment
- ✓ Limit blast radius
- ✓ Improve security

5. MULTI-ACCOUNT NETWORKING



- ✓ Connect multiple AWS accounts
- ✓ Centralized network management
- ✓ Secure cross-account access

6. LARGE-SCALE CONNECTIVITY



>>> ONE TRANSIT GATEWAY FOR ALL <<<

- ✓ High scalability
- ✓ Reduced complexity
- ✓ Lower operational overhead

7. PEERING vs TRANSIT GATEWAY

FEATURE	VPC PEERING	TRANSIT GATEWAY
Networks	2 VPCs only	Multiple VPCs / accounts
Transitivity	Non-transitive	Transitive (hub)
Scalability	Limited	High
Management	Manual	Centralized
Best use	VPC-to-VPC	Multi-VPC / Hybrid

- ✓ Use Peering for simple VPC-to-VPC.
- ✓ Use Transit Gateway for large-scale, multi-account, hub networking.





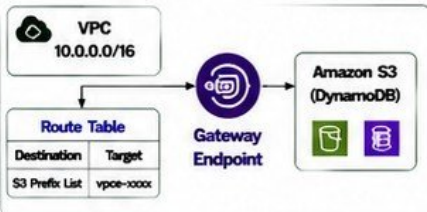
14. VPC ENDPOINTS AND AWS PRIVATELINK

PRIVATE, SECURE ACCESS TO AWS SERVICES AND YOUR OWN SERVICES

1. GATEWAY ENDPOINTS



- Connect your VPC to supported AWS services using your VPC route tables.
- No ENI, no hourly charges.
- Scales horizontally.
- Works at the network layer.



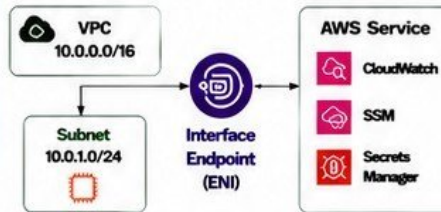
SUPPORTED SERVICES (EXAMPLES)

- ✓ Amazon S3
- ✓ Amazon S3 Glacier
- ✓ Amazon DynamoDB
- ✓ Amazon S3 Express One Zone

2. INTERFACE ENDPOINTS



- Create an ENI in your subnet with a private IP address.
- Access supported AWS services privately using Private IPs.
- Charged per AZ + data processing.
- Works at the application layer.



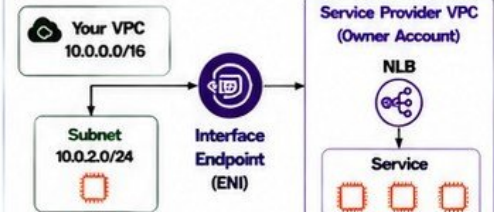
SUPPORTED SERVICES (EXAMPLES)

- ✓ Amazon CloudWatch
- ✓ AWS Secrets Manager
- ✓ AWS Systems Manager
- ✓ Amazon ECR API

3. AWS PRIVATELINK



- Access services hosted in another VPC, another account, or by AWS Partners.
- Uses Interface Endpoints.
- Secure, private, and highly available.
- No internet gateway or NAT required.



USE CASES

- ✓ Access SaaS services privately
- ✓ Cross-account service access
- ✓ Expose your services securely
- ✓ Partner service integrations

4. ENDPOINT POLICIES

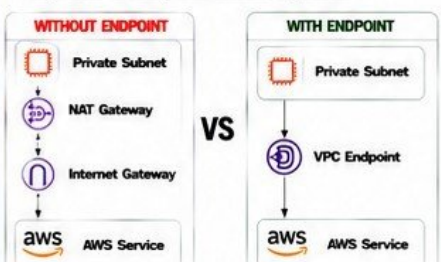
- Control access to services through endpoint policies.
- Acts as a firewall for your endpoint.
- Supports Allow and Deny statements.

EXAMPLE ENDPOINT POLICY (S3)

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::my-bucket/*"
    }
  ]
}
```

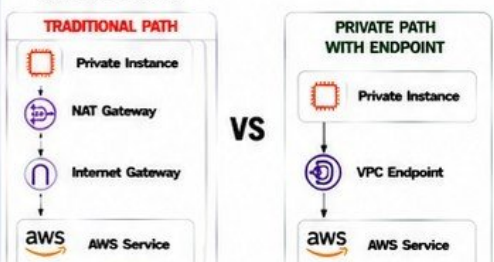
5. PRIVATE AWS SERVICE ACCESS

- Keep traffic on the AWS network.
- Access AWS services without internet, NAT, or public IPs.
- Improves security and reduces exposure.



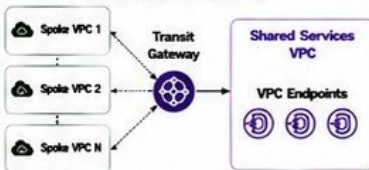
6. REMOVE UNNECESSARY INTERNET PATHS

- Endpoints eliminate the need for NAT Gateways for AWS service access.
- Reduces cost, simplifies architecture, and improves security.



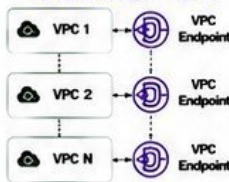
7. ENDPOINT ARCHITECTURE PATTERNS

A. CENTRALIZED ENDPOINTS (Shared Services VPC)



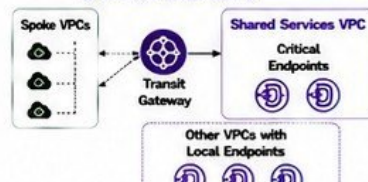
- Centralize endpoints in a shared VPC.
- Connect spoke VPCs via Transit Gateway.
- Simplifies management and control.

B. PER VPC ENDPOINTS



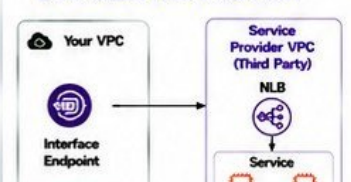
- Each VPC has its own endpoints.
- Simpler design for small environments.
- Higher cost at scale.

C. HYBRID ENDPOINTS



- Critical services centralized.
- Other services deployed locally.
- Balance between cost and control.

D. PRIVATELINK FOR THIRD-PARTY



- Use PrivateLink for SaaS or partner services.
- Private, secure, no internet exposure.
- Built on Interface Endpoints.

8. VPC PEERING vs TRANSIT GATEWAY vs ENDPOINTS vs PRIVATELINK

FEATURE	VPC PEERING	TRANSIT GATEWAY	VPC ENDPOINTS	AWS PRIVATELINK
Purpose	VPC-to-VPC connectivity	Hub-and-spoke networking	Private access to AWS services	Private access to services you own or third-party
Connectivity	VPC to VPC	Many VPCs via hub	VPC to AWS services	VPC to your/partner service
Transitive Routing	No	Yes	N/A	N/A
Cross-Account	No	No	Yes	No
Internet	Limited	High	High (per service)	High
Scale	Small connections	Large enterprise network	Access AWS services privately	Expose or consume private services
Use Case				

9. BEST PRACTICES

- ✓ Use Gateway Endpoints for S3 and DynamoDB to save cost.
- ✓ Use Interface Endpoints for other AWS services.
- ✓ Use PrivateLink for cross-account or third-party services.
- ✓ Apply endpoint policies to enforce least privilege.
- ✓ Centralize endpoints when managing many VPCs.
- ✓ Monitor endpoint usage with VPC Flow Logs.
- ✓ Remove unnecessary internet and NAT paths.
- ✓ Design for high availability across multiple AZs.
- ✓ Review and optimize endpoint strategy regularly.

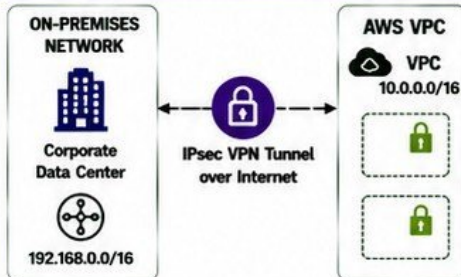




15. HYBRID AWS NETWORKING

CONNECT YOUR ON-PREMISES NETWORKS TO AWS SECURELY AND RELIABLY

1. SITE-TO-SITE VPN OVERVIEW



- ✓ Encrypted connection over the public internet.
- ✓ Uses IPsec VPN tunnels.
- ✓ Good for backup connectivity or small bandwidth needs.
- ✓ Supports dynamic routing (BGP) for route exchange.

2. VPN COMPONENTS

- VIRTUAL PRIVATE GATEWAY (VGW)**
The VPN concentrator on the AWS side. Attached to your VPC.
- CUSTOMER GATEWAY (CGW)**
Represents your on-premises network and VPN device.
- VPN CONNECTION**
The secure tunnel between VGW and CGW. Supports two tunnels for high availability.
- STATIC ROUTING (Optional)**
Use static routes instead of BGP if your device does not support BGP.

3. AWS DIRECT CONNECT



- ✓ Dedicated, private network connection to AWS.
- ✓ Higher bandwidth, lower latency, more consistent than VPN.
- ✓ Connects through AWS Direct Connect Locations.
- ✓ Supports both private VIF (VPC) and transit VIF (Transit Gateway).

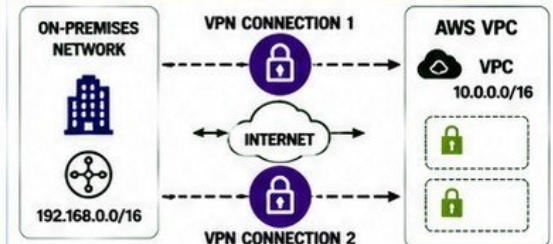
4. KEY COMPONENTS

- DIRECT CONNECT LOCATION (DX LOC)**
AWS facility where you establish your physical connection.
- DIRECT CONNECT ENDPOINT**
Logical interface in AWS that your connection uses.
- VIRTUAL INTERFACE (VIF)**
Private VIF connects to a VPC. Transit VIF connects to a Transit Gateway.
- BORDER GATEWAY PROTOCOL (BGP)**
Used to exchange routes dynamically between your router and AWS.

5. VPN vs DIRECT CONNECT

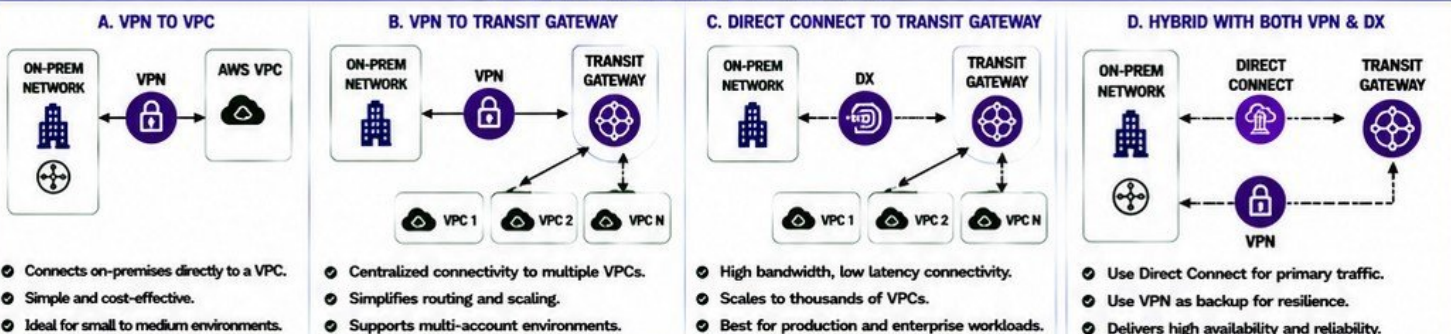
FEATURE	SITE-TO-SITE VPN	AWS DIRECT CONNECT
Connection Type	Over the public internet	Dedicated private connection
Bandwidth	Up to 1.25 Gbps per tunnel	From 50 Mbps up to 100 Gbps and higher
Latency	Higher and less consistent	Lower and more consistent
Availability	Good for backup connectivity	High availability production grade
Cost	Low	Higher
Setup Time	Minutes	Days to weeks
Use Case	Backup, dev/test, small offices	Production workloads, compliance, high bandwidth

6. REDUNDANT CONNECTIVITY



- ✓ Use two VPN tunnels for high availability.
- ✓ Tunnels must use different devices, ISPs, or physical paths.
- ✓ BGP helps reroute traffic automatically if one tunnel fails.

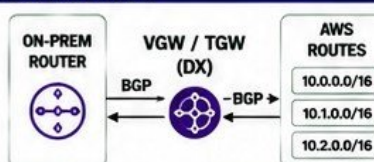
7. HYBRID ARCHITECTURE PATTERNS



- A. VPN TO VPC**
 - ✓ Connects on-premises directly to a VPC.
 - ✓ Simple and cost-effective.
 - ✓ Ideal for small to medium environments.
- B. VPN TO TRANSIT GATEWAY**
 - ✓ Centralized connectivity to multiple VPCs.
 - ✓ Simplifies routing and scaling.
 - ✓ Supports multi-account environments.
- C. DIRECT CONNECT TO TRANSIT GATEWAY**
 - ✓ High bandwidth, low latency connectivity.
 - ✓ Scales to thousands of VPCs.
 - ✓ Best for production and enterprise workloads.
- D. HYBRID WITH BOTH VPN & DX**
 - ✓ Use Direct Connect for primary traffic.
 - ✓ Use VPN as backup for resilience.
 - ✓ Delivers high availability and reliability.

8. ROUTING CONSIDERATIONS

- ✓ Use BGP for dynamic route exchange.
- ✓ Advertise only necessary routes.
- ✓ Avoid overlapping CIDR ranges.
- ✓ Use route summarization where possible.
- ✓ Validate routes using: show ip bgp summary
- ✓ Monitor route flaps and convergence.



9. BEST PRACTICES

- ✓ Use redundant connections (VPN or DX).
- ✓ Monitor VPN tunnels and DX connections.
- ✓ Use CloudWatch for VPN and DX metrics.
- ✓ Use Transit Gateway for scalability.
- ✓ Secure your on-premises devices.
- ✓ Document your network design and routes.



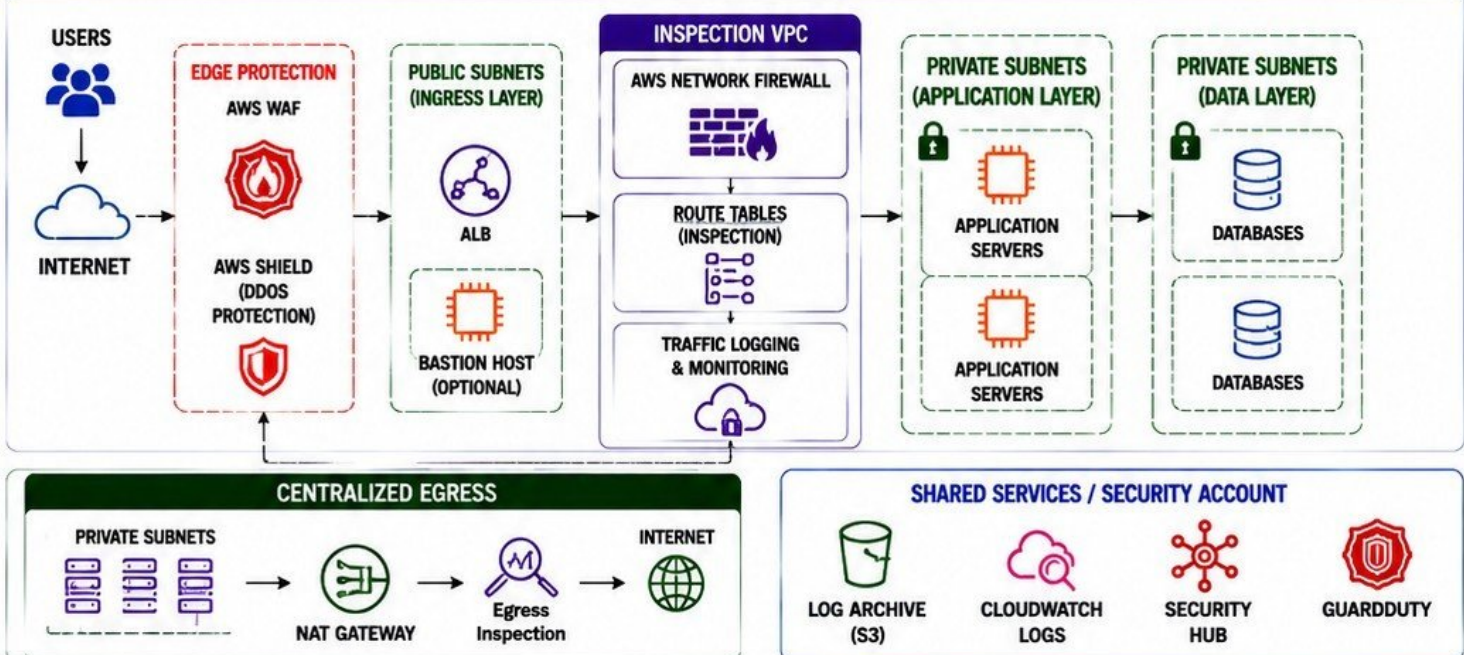


16. NETWORK SECURITY ARCHITECTURE

BUILD SECURE, RESILIENT AND SCALABLE AWS NETWORKS



TYPICAL SECURE NETWORK ARCHITECTURE



KEY SECURITY CONTROLS BY LAYER

LAYER	SECURITY CONTROLS
EDGE	AWS WAF, AWS Shield, Route 53 DNSSEC, TLS/SSL
NETWORK	VPC, Subnets, Route Tables, NACLs, Security Groups, VPC Flow Logs
INSPECTION	AWS Network Firewall, Traffic Mirroring, IDS/IPS, Threat Intelligence
APPLICATION	ALB Security, Private Subnets, Security Groups, Least Privilege Access
DATA	Private Subnets, NACLs, SGs, Encryption, Access Control
MONITORING	CloudWatch, VPC Flow Logs, GuardDuty, Security Hub, SIEM Integration

BEST PRACTICES

- ✓ Use least privilege for every access.
- ✓ Segment every tier and workload.
- ✓ Inspect all ingress and egress traffic.
- ✓ Centralize egress for visibility and control.
- ✓ Enable logging and monitoring everywhere.
- ✓ Apply zero-trust: verify explicitly, least access.
- ✓ Automate security across all accounts.
- ✓ Regularly review and test your security.

REMEMBER



Security is not a single product. It is an architecture and a process.

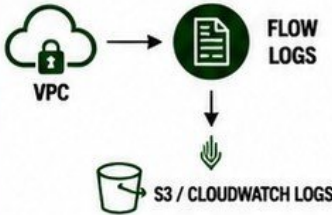




17. NETWORK MONITORING AND OBSERVABILITY

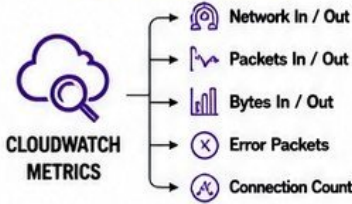
SEE EVERYTHING. UNDERSTAND EVERYTHING. FIX FASTER.

1 VPC FLOW LOGS



- Capture IP traffic metadata for ENIs in your VPC.
- See allowed and rejected traffic.
- Troubleshoot connectivity, security groups, and NACLs.
- Retention: S3 or CloudWatch Logs.

2 CLOUDWATCH NETWORK METRICS



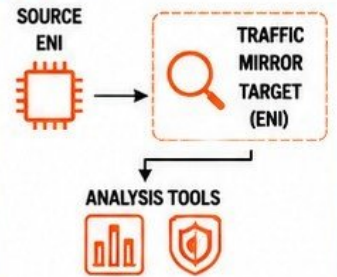
- Monitor network performance in real time.
- Per-ENI, per-VPC, per-ALB, per-NAT Gateway and more.
- Set alarms and dashboards.

3 ROUTE 53 RESOLVER QUERY LOGGING



- Log DNS queries from your VPCs.
- Identify DNS failures and unwanted domains.
- Improve security and visibility.

4 TRAFFIC MIRRORING



- Mirror live traffic for deep inspection.
- Use with tools like Wireshark, IDS/IPS, or packet analyzers.
- No performance impact on source workload.

5 REACHABILITY ANALYZER



- Validate network connectivity paths.
- Find routing, SG, NACL or network issues before they cause outages.
- Supports VPC, TGW, DX, VPN, Peering and more.

6 NETWORK ACCESS ANALYZER



- Identify resource access from external sources.
- Detect unintended public or cross-account access.
- Continuously monitor and validate access policies and rules.

7 BUILDING A TROUBLESHOOTING EVIDENCE TRAIL

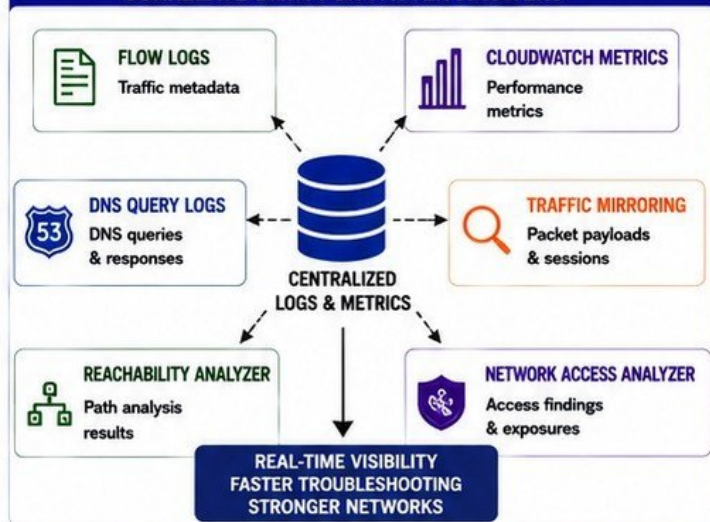


- Use multiple tools together for full visibility.
- Collect logs, metrics, traces and configurations.
- Document findings and resolutions.
- Build repeatable runbooks and dashboards.

WHERE EACH TOOL FITS

TOOL	PRIMARY PURPOSE	USE CASE EXAMPLES	DATA SOURCE
VPC FLOW LOGS	See IP traffic flow (allowed / rejected)	Why is traffic blocked? Which SG or NACL denied it?	ENI Traffic (L3-L4 metadata)
CLOUDWATCH NETWORK METRICS	Monitor performance and utilization	High network out? Packets dropped?	AWS Network Resources
ROUTE 53 RESOLVER QUERY LOGGING	Monitor DNS queries and failures	DNS resolution failures? Which domains queried?	DNS Queries (VPC Resolver)
TRAFFIC MIRRORING	Deep packet inspection and analysis	Inspect payloads, debug applications	Mirrored Packets (L3-L7)
REACHABILITY ANALYZER	Validate network paths	Can this resource reach that resource?	Network Topology & Configs
NETWORK ACCESS ANALYZER	Find unintended access	Is this resource publicly accessible?	Resource Policies & Configs

CORRELATE DATA FOR FASTER ANSWERS

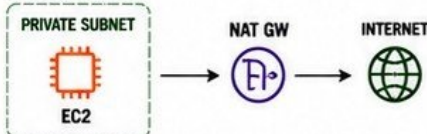




18. AWS NETWORK TROUBLESHOOTING

COMMON NETWORK ISSUES. ROOT CAUSES. HOW TO FIX THEM.

1 EC2 CANNOT REACH THE INTERNET



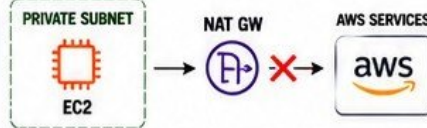
SYMPTOMS: No internet access, timeouts.

CHECK:

- Route table has 0.0.0.0/0 → NAT GW / IGW
- NAT Gateway is in a public subnet
- Internet Gateway attached to VPC
- Security Group allows outbound traffic
- NACL allows outbound traffic

FIX: Correct route, NAT/IGW setup, SG and NACL.

2 PRIVATE EC2 CANNOT REACH AWS SERVICES



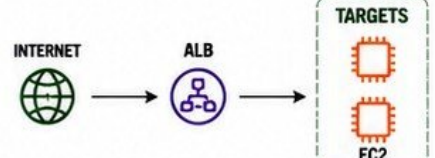
SYMPTOMS: Cannot reach S3, STS, ECR, etc.

CHECK:

- Route to NAT Gateway exists
- NAT Gateway has Elastic IP and is active
- Route table associated with subnet
- NACL allows outbound 443
- No endpoint for private access (optional)

FIX: Fix route/NAT or add VPC Endpoints.

3 ALB CANNOT REACH TARGETS



SYMPTOMS: 502 / 503 errors, targets unhealthy.

CHECK:

- Target group health check path/port
- Instances in correct subnets
- Security Group allows traffic from ALB SG
- NACL allows traffic from ALB subnets
- Instances running and listening on port

FIX: Fix health check, SG, NACL or target app.

4 VPCS CANNOT COMMUNICATE



SYMPTOMS: Cannot ping or connect.

CHECK:

- Peering/TGW connection exists and active
- Route tables in both VPCs updated
- Security Groups allow traffic
- NACLs allow traffic
- No overlapping CIDR blocks

FIX: Create connection, update routes, SG, NACL.

5 DNS RESOLUTION FAILURES



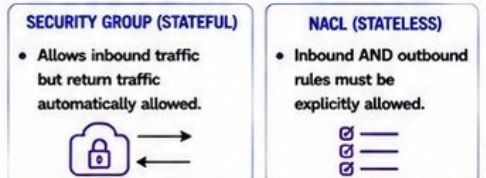
SYMPTOMS: Cannot resolve domain names.

CHECK:

- VPC DNS Hostnames enabled
- VPC DNS Resolution enabled
- Route 53 Resolver rules (if using on-prem DNS)
- Outbound access to internet (UDP/TCP 53)
- Security Group / NACL allows port 53

FIX: Enable DNS settings, fix routes/SG/NACL.

6 SECURITY GROUP vs NACL PROBLEMS



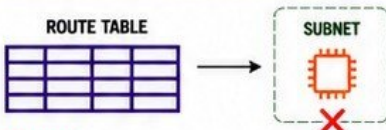
SYMPTOMS: Intermittent or one-way traffic.

CHECK:

- SG inbound rule exists
- SG allows outbound (default yes)
- NACL inbound rule exists
- NACL outbound rule exists
- Ephemeral ports (1024-65535) allowed

FIX: Add missing NACL rules or adjust rules.

7 INCORRECT ROUTE TABLES



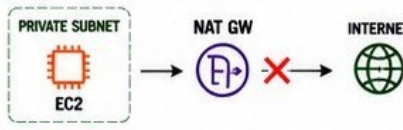
SYMPTOMS: Wrong destination, no connectivity.

CHECK:

- Route table associated with subnet
- Correct destination CIDR
- Next hop is valid (IGW, NAT, TGW, VPN, etc.)
- No conflicting routes
- Propagation status (for TGW / VGW)

FIX: Associate correct route table and fix routes.

8 NAT GATEWAY FAILURES



SYMPTOMS: Private instances lose internet.

CHECK:

- NAT Gateway status (available)
- NAT in a public subnet
- Elastic IP allocated
- Route table points to NAT Gateway
- NACL allows outbound
- NAT Gateway has sufficient capacity

FIX: Restore NAT, update route or recreate.

9 SYSTEMATIC TROUBLESHOOTING WORKFLOW

- 1 IDENTIFY THE PROBLEM**
Understand the symptom and user impact.
- 2 DETERMINE THE PATH**
Map the full network path end-to-end.
- 3 CHECK EACH COMPONENT**
Validate subnet, route table, SG, NACL, GW, endpoints along the path.
- 4 TEST AND COLLECT EVIDENCE**
Use logs, metrics, and packet tests.
- 5 ISOLATE ROOT CAUSE**
Find the exact component causing the issue.
- 6 FIX AND VALIDATE**
Apply fix and confirm resolution.

QUICK COMMANDS CHEAT SHEET

CHECK ROUTE TABLE

```
aws ec2 describe-route-tables \
--filters Name=association.subnet-id,
Values=subnet-xxxxxxxxxx
```

CHECK SG RULES

```
aws ec2 describe-security-groups \
--group-ids sg-xxxxxxxxxx
```

CHECK NACL ENTRIES

```
aws ec2 describe-network-acls \
--network-acl-ids acl-xxxxxxxxxx
```

CHECK NAT GATEWAY

```
aws ec2 describe-nat-gateways \
--nat-gateway-ids nat-xxxxxxxxxx
```

LOOKUP DNS

```
nslookup example.com
dig example.com
```



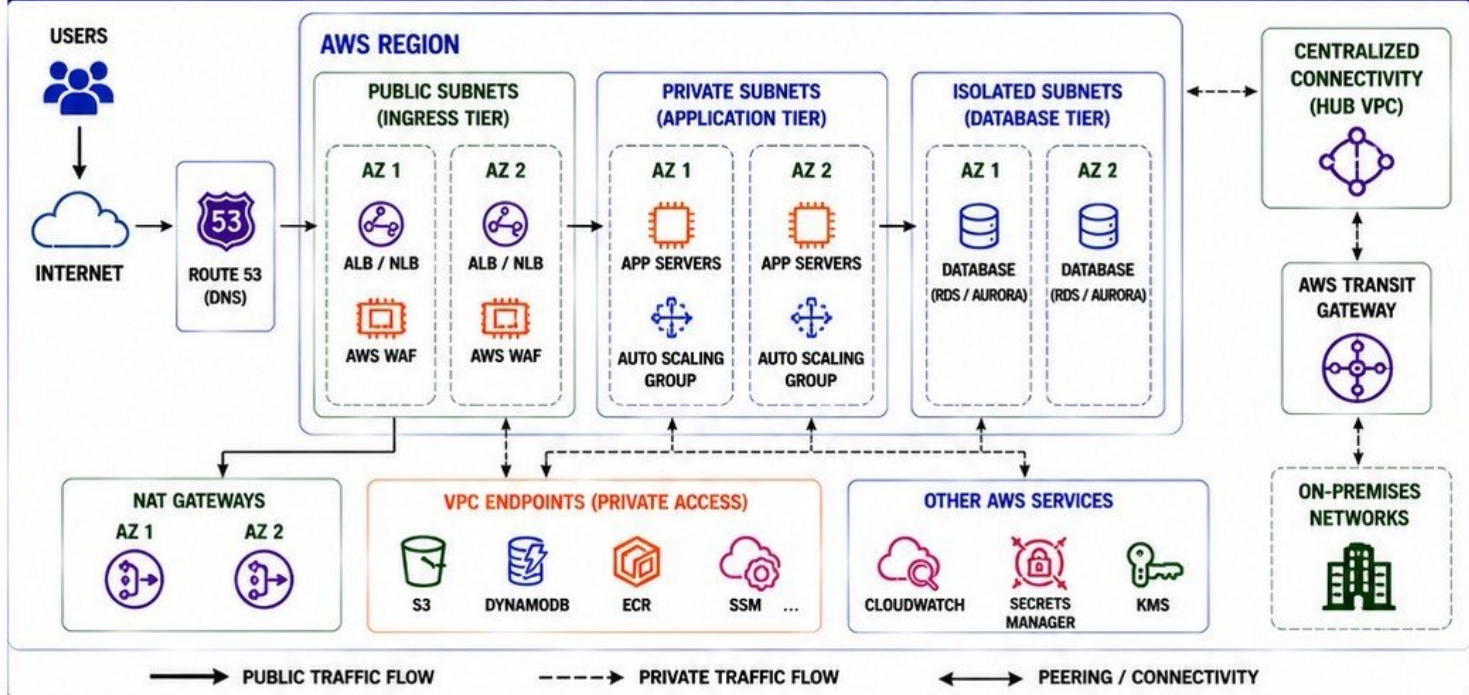


19. PRODUCTION AWS NETWORK ARCHITECTURE

HIGHLY AVAILABLE. SECURE. SCALABLE. WELL-ARCHITECTED.



1. HIGH-LEVEL ARCHITECTURE OVERVIEW



2. SECURITY BOUNDARIES

- ✓ PUBLIC SUBNETS: Only ALB/NLB and WAF.
- ✓ PRIVATE APP SUBNETS: No direct internet access.
- ✓ ISOLATED DB SUBNETS: No internet or public access.
- ✓ SGs: Stateful rules at instance level.
- ✓ NACLs: Stateless rules at subnet level.
- ✓ VPC FLOW LOGS: Enabled for all subnets.
- ✓ AWS WAF: Protects public endpoints.

3. HIGH AVAILABILITY DESIGN

- ✓ Multi-AZ deployment across at least 2 AZs.
- ✓ ALB/NLB spans multiple AZs.
- ✓ Auto Scaling Group across AZs.
- ✓ RDS Multi-AZ for automatic failover.
- ✓ NAT Gateways in each public subnet.
- ✓ No single points of failure.
- ✓ Health checks and failover built-in.

4. KEY COMPONENTS & PURPOSE

COMPONENT	PURPOSE
ALB / NLB	Distribute traffic to application tier
NAT GATEWAY	Allow private instances internet access
VPC ENDPOINTS	Private access to AWS services
TRANSIT GATEWAY	Connect VPCs and on-premises networks
ROUTE 53	DNS resolution and routing
WAF	Protect web applications from threats

5. NETWORK TRAFFIC FLOW

- 1 Users resolve domain via Route 53.
- 2 Traffic goes to ALB/NLB in public subnets.
- 3 ALB forwards traffic to app servers in private subnets.
- 4 App servers access databases in isolated subnets.
- 5 Private subnets use NAT Gateway for internet.
- 6 Private resources use VPC Endpoints for AWS services.
- 7 On-premises connects via Transit Gateway.

6. HIGH AVAILABILITY & FAILURE PLANNING

- ✓ AZ failure: Traffic rerouted automatically by ALB and ASG.
- ✓ NAT Gateway failure: Traffic uses NAT in another AZ.
- ✓ Instance failure: ASG replaces unhealthy instances.
- ✓ Database failure: RDS Multi-AZ automatic failover.
- ✓ Network failure: Redundant paths via TGW and AZs.
- ✓ Regular DR testing and documented runbooks.

7. BEST PRACTICES

- ✓ Use least privilege security rules.
- ✓ Segment workloads using subnets.
- ✓ Use VPC Endpoints instead of NAT where possible.
- ✓ Enable monitoring, logging and alerting.
- ✓ Continuously review routes, rules and endpoints.
- ✓ Document your network architecture.





20. ENTERPRISE BEST PRACTICES AND FINAL REVIEW

DESIGN SECURE. BUILD RESILIENT. OPERATE RELIABLY. OPTIMIZE CONTINUOUSLY.

1. PRODUCTION NETWORK CHECKLIST



- ✓ Multi-AZ design for all critical tiers
- ✓ Public subnets isolated from data
- ✓ Private subnets for application workloads
- ✓ Isolated subnets for databases
- ✓ Least privilege Security Groups
- ✓ NACLs used for subnet-level protection
- ✓ Route tables associated to correct subnets
- ✓ NAT Gateway in each AZ for HA
- ✓ VPC Endpoints for AWS service access
- ✓ Flow Logs enabled for all VPCs
- ✓ DNS logging enabled (Route 53 Resolver)
- ✓ Monitoring, alerts and dashboards in place
- ✓ Backup connectivity (VPN or Direct Connect)
- ✓ Disaster recovery tested and documented

2. CIDR PLANNING STANDARDS



Use private RFC1918 address space.

- 10.0.0.0/8 (Large environments)
- 172.16.0.0/12 (Medium environments)
- 192.168.0.0/16 (Small environments)

BEST PRACTICES

- Plan for growth, not just today.
- Allocate larger CIDRs to VPCs first.
- Use /20 or larger for VPCs.
- Use /24 for subnets (minimum).
- Keep CIDR blocks non-overlapping.
- Reserve CIDRs for future accounts.
- Document CIDR usage and ownership.

3. HIGH-AVAILABILITY PRINCIPLES



- ✓ Deploy across at least 2 AZs.
- ✓ No single point of failure.
- ✓ Use ALB/NLB across multiple AZs.
- ✓ Use Auto Scaling for stateless tiers.
- ✓ Use Multi-AZ RDS or Aurora.
- ✓ Use NAT Gateway in every AZ.
- ✓ Route 53 health checks and failover.
- ✓ Use redundant connectivity (VPN + Direct Connect or Multi-VPN).
- ✓ Test failure scenarios regularly.
- ✓ Design for failure, not for success.

4. SECURITY RECOMMENDATIONS



- ✓ Follow least privilege everywhere.
- ✓ Deny ALL by default (NACLs).
- ✓ Allow by exception (Security Groups).
- ✓ Use Private Subnets for all workloads.
- ✓ Use AWS Network Firewall for inspection.
- ✓ Use AWS WAF for web applications.
- ✓ Encrypt data in transit (TLS) and at rest.
- ✓ Use VPC Endpoints to avoid internet access.
- ✓ Enable DNS Firewall and DNSSEC.
- ✓ Enable Flow Logs and log aggregation.
- ✓ Centralize security logging and alerting.
- ✓ Regularly review access and rules.

5. COST CONSIDERATIONS



- ✓ NAT Gateway charges are per hour and per GB.
- ✓ Use NAT Gateway per AZ (not cross-AZ).
- ✓ Use VPC Endpoints to reduce data transfer costs.
- ✓ Avoid inter-AZ data transfer.
- ✓ Right-size CIDR blocks to avoid waste.
- ✓ Use PrivateLink for third-party services when possible.
- ✓ Monitor with Cost Explorer, Budgets and Alerts.

6. COMMON ARCHITECTURE MISTAKES



- ✗ All resources in public subnets.
- ✗ Single NAT Gateway for all AZs.
- ✗ Security Groups too open (0.0.0.0/0).
- ✗ Missing Flow Logs and monitoring.
- ✗ Using IGW for private workloads.
- ✗ No VPC Endpoints for AWS services.
- ✗ Overlapping CIDR blocks.
- ✗ No HA design (single AZ).
- ✗ Not testing failover and recovery.
- ✗ Hardcoded IPs and manual routing.

7. AWS NETWORKING TROUBLESHOOTING CHECKLIST



CONNECTIVITY

- ✓ Can the source reach the destination?
- ✓ Check route tables.
- ✓ Check NACLs.
- ✓ Check Security Groups.
- ✓ Check Network ACL order.



DNS

- ✓ Check DNS resolution.
- ✓ Check Route 53 Resolver rules.
- ✓ Check DNS query logs.
- ✓ Check VPC DNS settings.
- ✓ Verify domain / records.



ROUTING

- ✓ Correct route table associated?
- ✓ Any more specific routes?
- ✓ Blackhole routes?
- ✓ Check TGW / Peering routes.



TRAFFIC INSPECTION

- ✓ Is traffic passing through Firewall / WAF?
- ✓ Check rule groups.
- ✓ Check logs for denials.
- ✓ Check TLS / certificates.
- ✓ Check endpoint policies.



OBSERVABILITY

- ✓ Check VPC Flow Logs.
- ✓ Check CloudWatch metrics.
- ✓ Check Reachability Analyzer.
- ✓ Check Network Access Analyzer.



ENDPOINTS

- ✓ Is the service using correct endpoint?
- ✓ Check SG on endpoint.
- ✓ Check DNS name (vpce-xxxx).
- ✓ Check endpoint policy.

8. ARCHITECTURE DECISION GUIDE

DECISION AREA	OPTIONS	WHEN TO CHOOSE
PUBLIC ACCESS	ALB + WAF	Web applications, user traffic
PRIVATE ACCESS	PrivateLink / VPC Endpoint	SaaS / AWS private access
HYBRID CONNECTIVITY	VPN / DX / Transit Gateway	On-premises integration
INSPECTION	Network Firewall / GWLB	Deep packet inspection
EGRESS	NAT Gateway / VPC Endpoints	Internet / AWS service access
DNS	Route 53 Resolver / DNS Firewall	Enterprise DNS management

9. SENIOR ENGINEER RECOMMENDATIONS



- ✓ Design for failure, not for success.
- ✓ Automate everything (IaC).
- ✓ Document architecture decisions.
- ✓ Monitor before users complain.
- ✓ Keep security simple and effective.
- ✓ Test regularly: failover, DR, and chaos engineering.
- ✓ Use clear naming, tagging and standards.
- ✓ Review and refactor continuously.
- ✓ Think long-term: scale, cost, security, operability.
- ✓ Make the next engineer successful.

10. FINAL REVIEW SUMMARY



A GREAT NETWORK IS:

- ✓ SECURE BY DESIGN
- ✓ HIGHLY AVAILABLE
- ✓ WELL ARCHITECTED
- ✓ COST OPTIMIZED
- ✓ EASY TO OPERATE
- ✓ READY FOR SCALE

 BUILD IT RIGHT.
 RUN IT WELL. SCALE IT SAFELY.
VERI QTA
